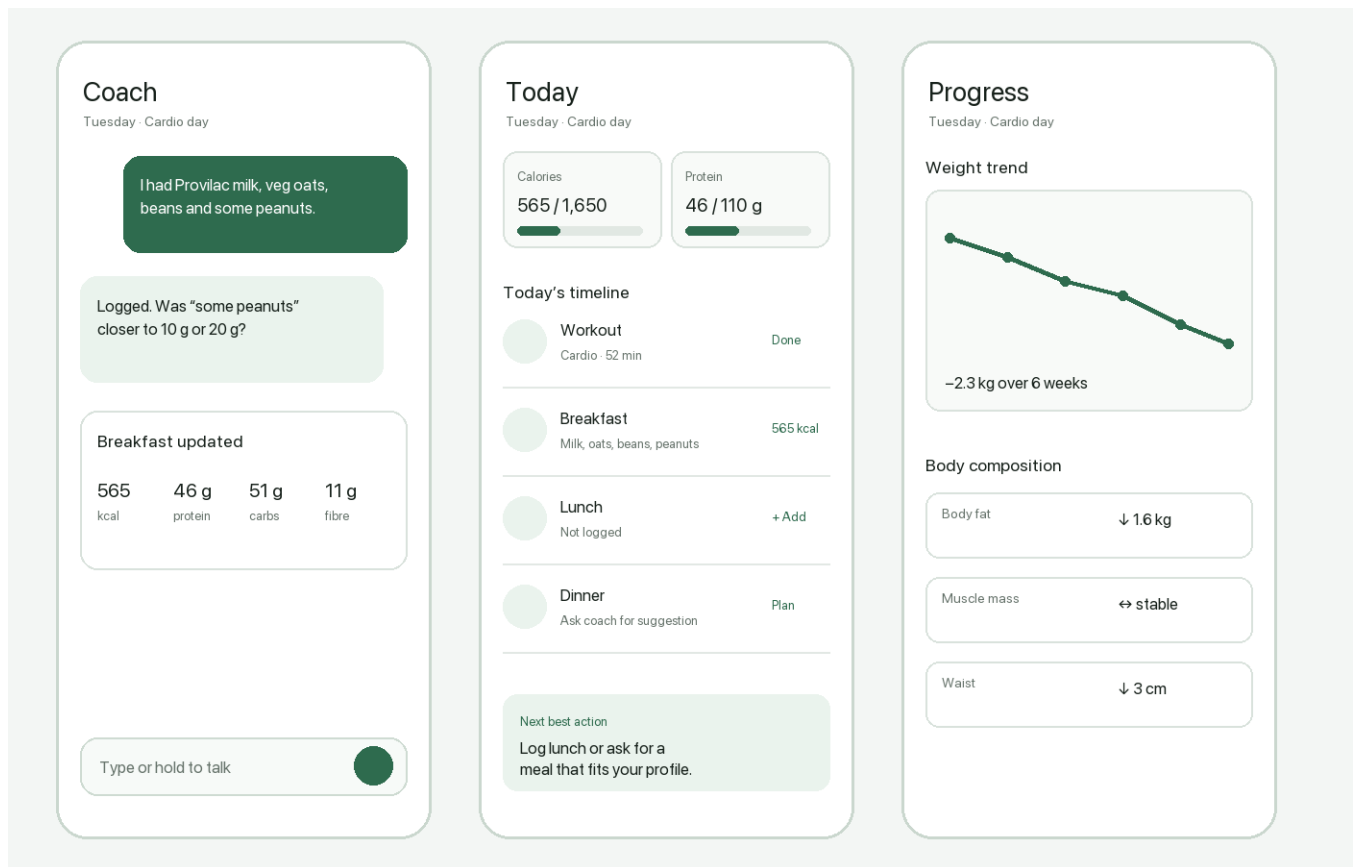


Conversational Nutrition and Fitness App

Product Requirements and Experience Design

NutriTrainer AI

Offline first • India-wide personalization • Voice and chat first



Version 1.0 • 8 September 2026

Document purpose

This document defines a buildable product for a conversational nutrition and fitness companion. It converts natural voice or chat into structured meal, exercise, weight and body composition records; calculates nutrition through deterministic software; and provides culturally relevant guidance without requiring laboratory-level input.

The recommended first release uses Kotlin Multiplatform and Compose Multiplatform, encrypted local storage, platform on-device models where available and a replaceable AI engine. The food taxonomy is India-wide from the start, while curated regional content can ship in independently versioned packs.

Primary decision

Conversation and the Today tracker are two synchronized views of one record. AI interprets language and explains results. It does not own calories, BMI, BCA calculations, recipe arithmetic, history or final recommendation rules.

Item	Decision
Platforms	Android and iOS.
Audience	Adults tracking food, exercise, weight and body composition.
Clinical scope	Wellness and nutrition guidance, not diagnosis or treatment.
Account requirement	Core offline use requires no account or external AI subscription.
Product name	NutriTrainer AI. Tagline: Your personal nutrition and fitness trainer.

Contents

- Vision and users
- Experience and screen design
- Detailed requirements
- Nutrition confidence and data sources
- Local AI and technology architecture
- Privacy, safety and accessibility
- Testing, rollout and risks

1 Product vision and success

People should describe what they ate and how they exercised in the language they naturally use. The application quietly turns that conversation into an editable daily record and offers one practical next action.

Problems to solve

- Traditional trackers require repetitive search, weighing and forms.
- Indian household foods vary by recipe, water, oil, thickness and serving size.
- Generic databases have weak coverage for Indian regional dishes, local-language aliases, transliterations and brands.
- Chat-only tracking can hide what was saved and confuse planned meals with consumed meals.
- Cloud-first AI adds cost, privacy and connectivity dependencies.

Goals and targets

Goal	Initial target
Low effort	Common meal logged by voice or chat in under 20 seconds.
Trust	Every item is editable and shows source, serving basis and confidence.
Cultural relevance	All states and UTs searchable; launch packs meet published coverage and quality thresholds.
Offline utility	Logging, calculations, history and Today work without network.
Interpretation	90% of logs accepted or corrected successfully; improvement after calibration.
Reliability	99.5% crash-free sessions.

Non goals

- Clinical diagnosis or treatment
- Perfect calorie measurement from casual descriptions
- Continuous on-device model training
- A social feed
- Full restaurant-menu coverage
- Replacing a dietitian or physician

2 Users and jobs

Persona	Needs	Product response
Routine tracker	Fast daily logging	Learn usual foods, one-tap repeat and voice input.
Fat-loss learner	Trends without shame	Ranges, rolling averages and practical guidance.

Persona	Needs	Product response
Strength trainee	Protein and recovery context	Workout-aware remaining targets.
Regional household eater	Home recipes absent from generic databases	Local aliases, family recipes and calibration.
Data-conscious user	Weight, BCA and wearable data	Structured comparisons and progress charts.
Low-connectivity user	Reliable offline access	Local database, calculations and parser fallback.

Jobs to be done

- Log a meal naturally before I forget.
- Estimate uncertain portions without forcing daily weighing.
- Remember my household version but allow exceptions.
- Project a planned meal without counting it as consumed.
- Replace incorrect records when I correct the assistant.
- Suggest a familiar next meal using today's remaining needs.
- Explain weight trends without overreacting to one measurement.
- Compare compatible BCA readings and disclose uncertainty.

3 Experience principles and navigation

Principle	Required behaviour
Conversation first	One composer accepts voice, text, photo and barcode.
Structured underneath	Every result becomes a visible editable record.
Approximate honestly	Use ranges when inputs are ambiguous.
Ask selectively	Ask one question only when it changes advice materially.
Learn with consent	Offer to save defaults; never redefine food silently.
Regional by design	Community guides ranking without stereotyping.
Calm guidance	No guilt or alarm for normal variation.
Graceful degradation	No network or LLM must not block core tracking.

Navigation

Tab	Purpose	Primary actions
Coach	Conversational entry and guidance	Talk, type, attach, scan, confirm, correct and ask.
Today	Authoritative daily record	Review totals, meals, exercise, plans and edits.
Progress	Trends	Weight, BMI, BCA, waist, nutrition and workouts.
Library	Foods and recipes	Search, verify, create household variants and favourites.
Profile	Goals and controls	Targets, community, diet, permissions, backup and AI.

Screen concept

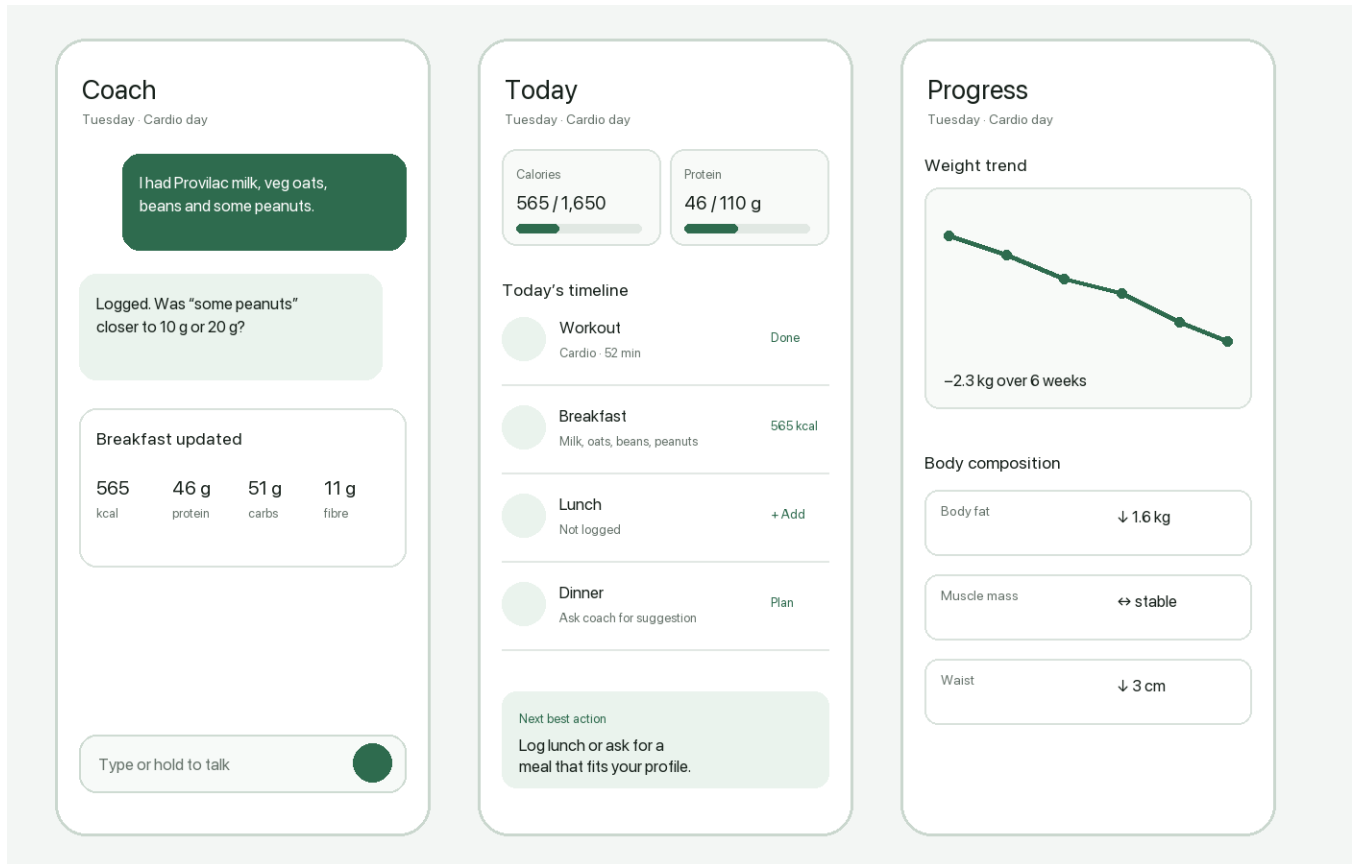


Figure 1 Coach, Today and Progress are connected views of the same local data.

4 Detailed screen design

4.1 Onboarding

Step	Content	Policy
Welcome	Explain conversational logging, editable estimates, privacy and offline use; collect no answer.	Informational only; Get started continues.
Profile	Age band, optional sex for equations, height, units and weight.	Weight can be skipped.
Goal	Maintain, lose fat, gain muscle or weight, lose fat plus gain muscle, improve nutrition or track only.	One primary goal; body recomposition is a distinct option.
Food category	Vegetarian; veg plus non-veg; vegan; or veg plus eggs (eggetarian).	Required safety choice; non-veg always includes vegetarian foods.
Category details	Multi-select non-veg foods eaten; dairy/egg handling, fasting and hard exclusions.	Conditional multi-select; never implies every meal contains non-veg.
Region	Country, state/UT and optional sub-region or city.	Searchable; suggested from device locale, never preselected.
Food culture	Optional community or cultural food tradition, home cuisine and other cuisines enjoyed.	Optional, multi-select, editable and skippable.
Meal routine	Egg or non-veg days, fasting days, staple grains, usual meal times and foods to avoid.	Optional; can be learned later.
Language mode	English only, English plus one or more regional languages, or regional language only.	Searchable multi-language configuration; app UI and logging languages stored separately.
AI setup	System AI availability and future optional model download.	Basic mode always works.
Permissions	Microphone, camera, health and notifications.	Request only in context; all optional.

Regional and dietary personalization rules

- Food category is the primary classification and safety filter; region and culture personalize only within the allowed category.
- Vegetarian excludes meat, poultry, fish and eggs; vegan also excludes dairy, honey and other animal-derived foods; veg plus eggs allows vegetarian meals and eggs; veg plus non-veg includes the complete vegetarian catalogue plus only the animal foods explicitly selected by the user.
- Selecting veg plus non-veg never means every meal or recommendation should contain meat or fish. The ranking engine should freely recommend vegetarian meals and learn the user's actual frequency.
- India has no default or first cuisine. Device locale may suggest a state or language, but the user must confirm it; location permission is not required.
- Region ranks relevant names, serving units and dishes. Community or cultural tradition is optional, searchable and never used to infer religion, caste or dietary restrictions.

- Users may select several food cultures and cuisines, such as Tamil home food with Chettinad and Kerala favourites, or Marathi home food with Gujarati favourites.
- Household choices override broad regional patterns. A Tamil vegetarian, Marathi veg-plus-non-veg, Gujarati veg-plus-non-veg or Punjabi vegan must each receive suitable options.
- Temporary day-level modes such as fasting, no-egg day or vegetarian today override the normal pattern without changing the permanent profile.
- Allergy, medical and explicitly stated religious exclusions outrank cuisine and recommendation preferences.

Priority order	Example
1 Safety exclusions	Allergy or clinical restriction.
2 Current-day rule	Fasting, vegan today or no-egg day.
3 Primary food pattern	Vegetarian, veg plus non-veg, vegan or veg plus eggs.
4 Multi-select food details	Animal foods eaten and household exclusions.
5 Household defaults	Usual milk, roti or rice, recipes and portions.
6 State, sub-region and food culture	Tamil, Marathi, Gujarati, Bengali, Naga and other ranking signals.
7 Additional cuisines	Foods enjoyed outside the home cuisine.

4.1.1 India-wide regional coverage

The catalogue must cover all Indian states and union territories rather than treating a few large cuisines as national defaults. State is an indexing aid, not a claim that every household in that state eats alike.

Zone	State / union territory options
North	Chandigarh; Delhi (NCT); Haryana; Himachal Pradesh; Jammu and Kashmir; Ladakh; Punjab; Rajasthan; Uttar Pradesh; Uttarakhand.
West and Central	Chhattisgarh; Dadra and Nagar Haveli and Daman and Diu; Goa; Gujarat; Madhya Pradesh; Maharashtra.
South	Andaman and Nicobar Islands; Andhra Pradesh; Karnataka; Kerala; Lakshadweep; Puducherry; Tamil Nadu; Telangana.
East	Bihar; Jharkhand; Odisha; West Bengal.
Northeast	Arunachal Pradesh; Assam; Manipur; Meghalaya; Mizoram; Nagaland; Sikkim; Tripura.

Sub-regional and food-tradition examples

Area	Examples for searchable tags—not a closed list
Tamil Nadu	Tamil; Chettinad; Kongunadu; Nanjil Nadu; Madurai; coastal Tamil.
Maharashtra	Marathi; Maharashtrian; Konkan; Malvani; Vidarbha; Khandesh; Kolhapuri; Pune.
Gujarat	Gujarati; Kathiyawadi; Surti; Kutchi; Charotari; Jain-style household food.
Karnataka / Kerala	Udupi; Mangalorean; Kodava; North Karnataka; Kerala; Malabar; Syrian Christian; Moplah.
Andhra Pradesh / Telangana	Andhra; Rayalaseema; coastal Andhra; Telangana; Hyderabad.
North and Central	Punjabi; Dogri; Kashmiri; Himachali; Garhwali; Kumaoni; Awadhi; Bhojpuri; Braj; Rajasthani; Marwari; Malwi; Bundelkhandi; Chhattisgarhi.
East	Bengali; Odia; Bihari; Maithil; Magahi; Jharkhand regional; Sikkimese.
Northeast	Assamese; Naga; Mizo; Khasi; Garo; Manipuri/Meitei; Tripuri; Arunachali traditions.
Pan-India / heritage	Mughlai; Parsi; Sindhi; Bohri; Konkani; Anglo-Indian; Tibetan; Indo-Chinese.

Implementation: ship state/UT tags in the base pack; add curated sub-regional packs over time; support aliases in local scripts; and always provide Search, Other / describe mine, Multiple traditions and Skip. New community labels require editorial review and must not automatically assign diet rules.

4.1.2 Developer-ready onboarding flow

Use a single-task screen for each decision, a visible progress indicator and a persistent Back action. Save each completed answer locally as a draft so app termination never forces a restart. Only the food pattern is mandatory for dietary personalization; every culture field can be skipped.

ID and screen	UI and interaction	Continue rule / stored result
OB-01 Welcome / value	No question and no selectable cards. Show logo, three benefit statements, a local-data privacy line and Get started; Sign in is absent from the offline-first MVP.	Tap Get started; onboardingVersion.
OB-02 Food pattern	Four single-select cards: Vegetarian; Veg + non-veg; Vegan; Veg + eggs (Eggetarian). Explain that veg + non-veg includes all vegetarian food and does not imply non-veg at every meal.	Exactly one required; foodPattern.
OB-03 Food details	Multi-select check cards. Veg + non-veg: eggs, chicken, mutton/goat, fish/seafood, pork, beef/buffalo and other. Vegetarian/eggetarian: dairy choices. Vegan: optional ingredients the user also avoids.	Multiple selections allowed; allowedAnimalFoods, dairyAllowed and foodDetailIds.
OB-04 Safety exclusions	Searchable allergy list plus None and Add my own. Separate “foods I avoid” field; do not combine preference with allergy.	None or one selection required; allergyIds and avoidanceText.
OB-05 State or UT	Search field and grouped full India list. Offer “Use device language suggestion” and “I live outside India”. Never silently select Gujarat or any state.	Optional but recommended; countryCode and stateUtCode.
OB-06 Food traditions	Searchable multi-select tags based on the chosen state, plus Browse all India, Multiple traditions, Describe mine and Skip. Explain that this improves names and suggestions only.	Zero or more; traditionIds and customTraditionText.
OB-07 Home food profile	Select staple grains/breads, usual oils, dairy type, spice level, meal schedule and optional fasting pattern. “Teach later in chat” is available.	Optional; HouseholdFoodProfile draft.
OB-08 Goal and body profile	Single-select goal cards: maintain; lose fat; gain muscle/weight; lose fat + gain muscle (body recomposition); improve nutrition/protein; track only. Then units, age band, height and optional weight.	One goal required; validate plausible ranges; primaryGoal and bodyProfile.
OB-09 Voice and language	First choose app display language. Then choose logging mode: English only; English + one or more regional languages; regional language only; or another combination. Regional languages use a searchable multi-select. Request microphone permission only after Tap to try.	Skippable; appLocale, loggingLanguageCodes[] and permission result.
OB-10 Review	Editable summary: category first, then exclusions, region/traditions, household choices and goal. Start tracking button.	Commit one transaction; onboardingStatus=complete.

Branching and back-navigation

- OB-02 → OB-03 always; show only fields relevant to the selected primary category.
- Changing category later immediately revalidates saved animal-food details and flags incompatible household foods for confirmation.
- Back preserves answers. Skip stores an explicit skipped state, not an empty value that could be mistaken for loading failure.
- Deep links into onboarding resume at the first incomplete required screen. Review can edit any section and return without losing later answers.

4.1.3 Onboarding visual and interaction specification

Element	Developer specification
Layout	Safe-area column; 24 dp horizontal padding; progress and Back at top; title, helper text and content scroll independently; Continue pinned above the bottom safe area.
Progress	Text “Step n of 10” plus a thin determinate bar. Conditional screens count once activated; announce progress to screen readers.
Choice cards	Minimum 56 dp height, 16 dp radius and 12 dp gap. Entire card is tappable. Selected state uses green border, pale-green fill and checkmark; do not rely on colour alone.
Primary category screen	Two-column grid on widths ≥ 360 dp, one column below 360 dp or at large text sizes. Keep all four choices visible before secondary settings.
Search and tags	Search matches English, local script and transliteration. Selected tags appear above results with removable controls. Results include state/sub-region context to resolve duplicate names.
Buttons	Primary: Continue / Start tracking. Secondary: Skip for now. Back is never destructive. Disabled Continue includes accessible reason.
Copy tone	Neutral and specific: “What kind of food do you eat?” Avoid “pure veg”, moral labels, assumptions and guilt.
Offline behaviour	All base choices and the full state/UT index work offline. If a regional pack is unavailable, save the preference and use the national catalogue until download.
Loading/error	Skeleton only for downloadable suggestions. Base list must not load remotely. Error offers Retry and Continue with basic catalogue.
Accessibility	44×44 pt iOS / 48×48 dp Android targets, dynamic type, logical focus, labels for icons, reduced motion and WCAG AA contrast.
Responsive	Phone portrait is canonical. Tablet limits content to 600 dp and keeps bottom action within that column.

Canonical screen copy

Screen	Title	Helper / action
Food pattern	Which foods do you eat?	Veg + non-veg includes vegetarian meals too; it does not mean every meal is non-veg. · Continue
Region	Which region's food feels like home?	This helps us recognise local names and portions. It does not limit what you can eat. · Skip for now
Food traditions	Any food traditions you follow?	Choose as many as you like. We never assume your diet from your community. · Continue
Review	Does this look right?	Your food category always takes priority over regional suggestions. · Start tracking

4.1.4 Onboarding screen designs

These high-fidelity wireframes are the canonical visual reference for hierarchy, selection states, navigation and copy placement. Product artwork and final icons may change without changing the interaction contract.

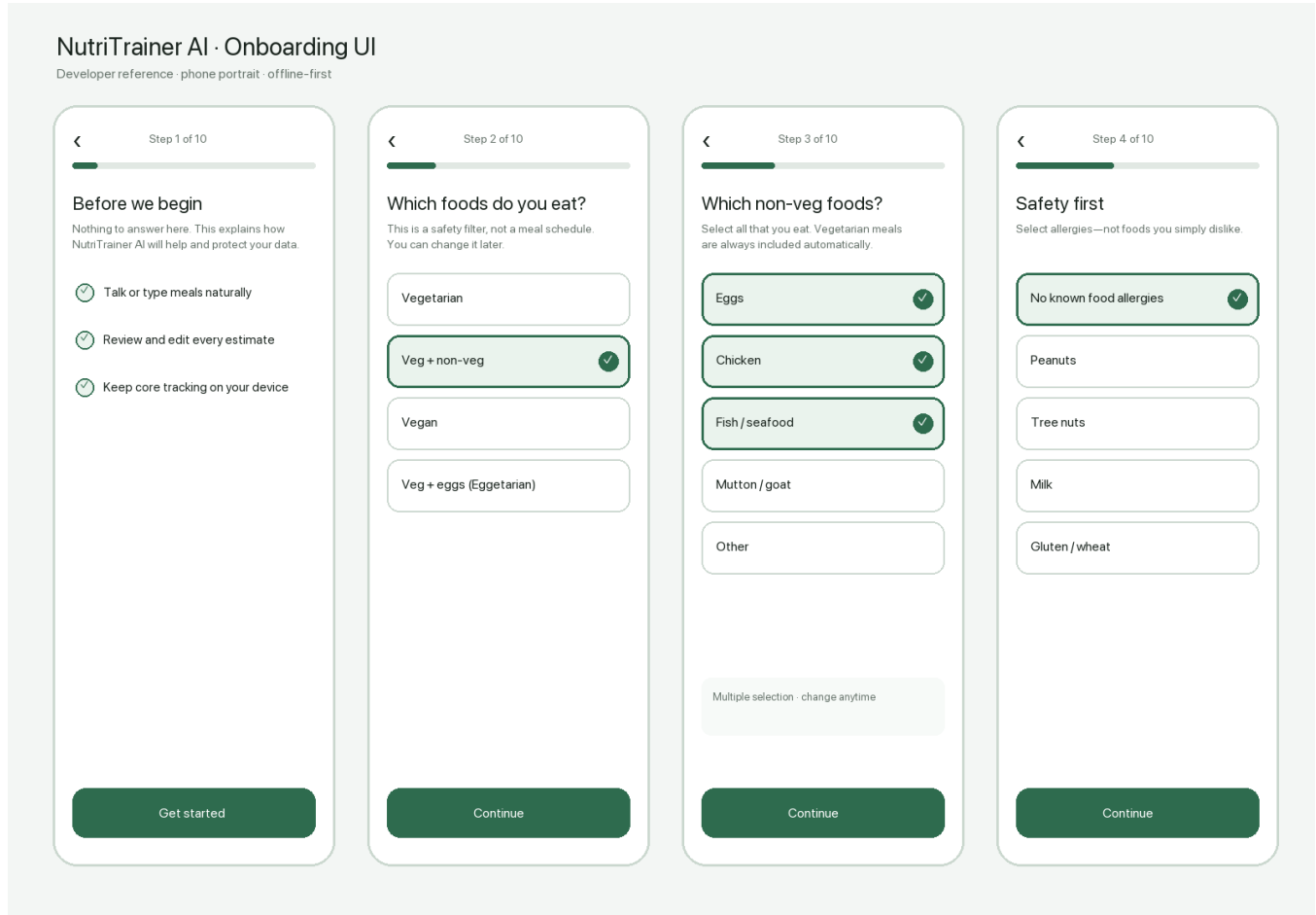


Figure 2 Informational welcome, food pattern, multi-select food details and allergy separation.

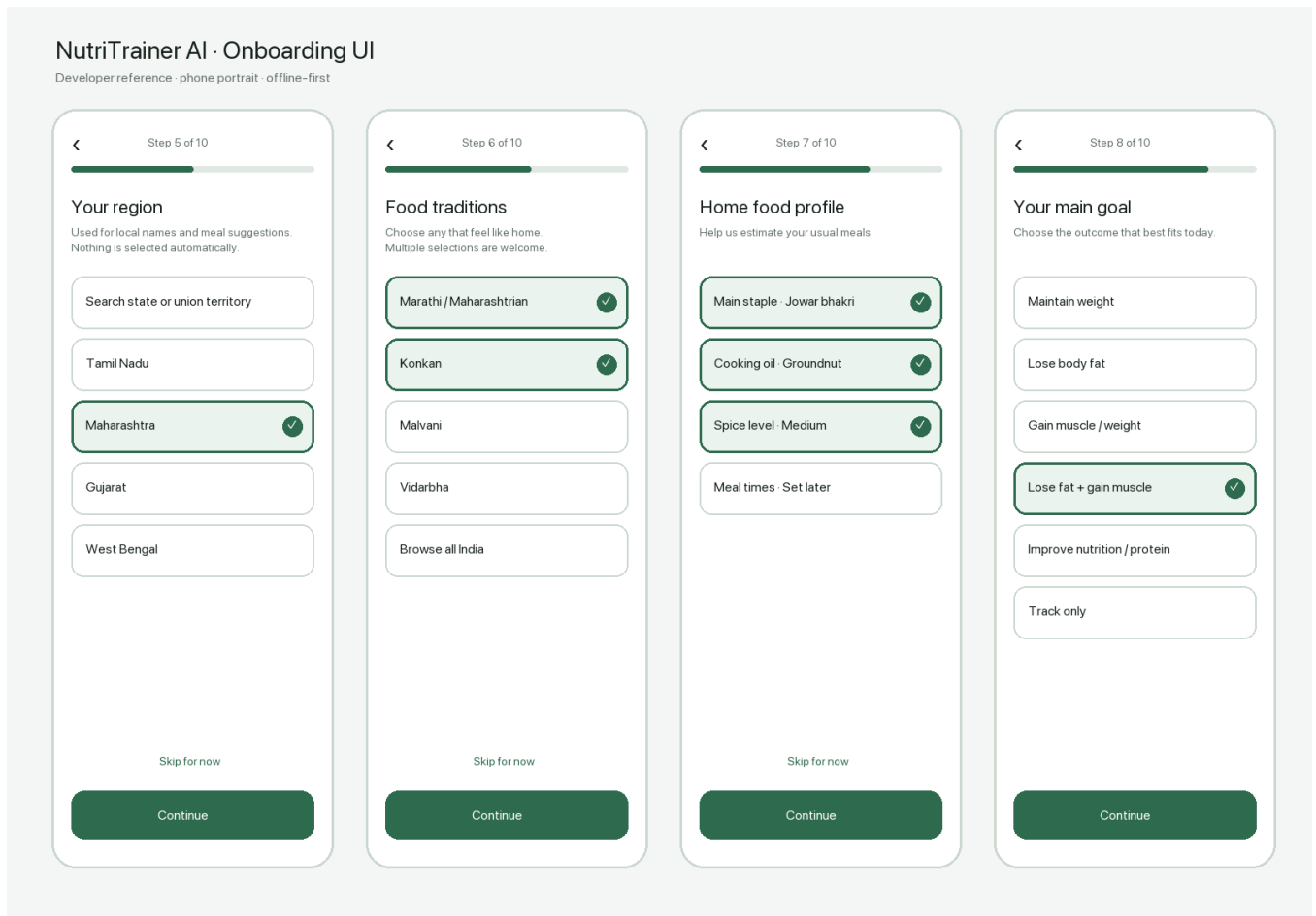


Figure 3 State/UT, optional food traditions, household profile and goal.

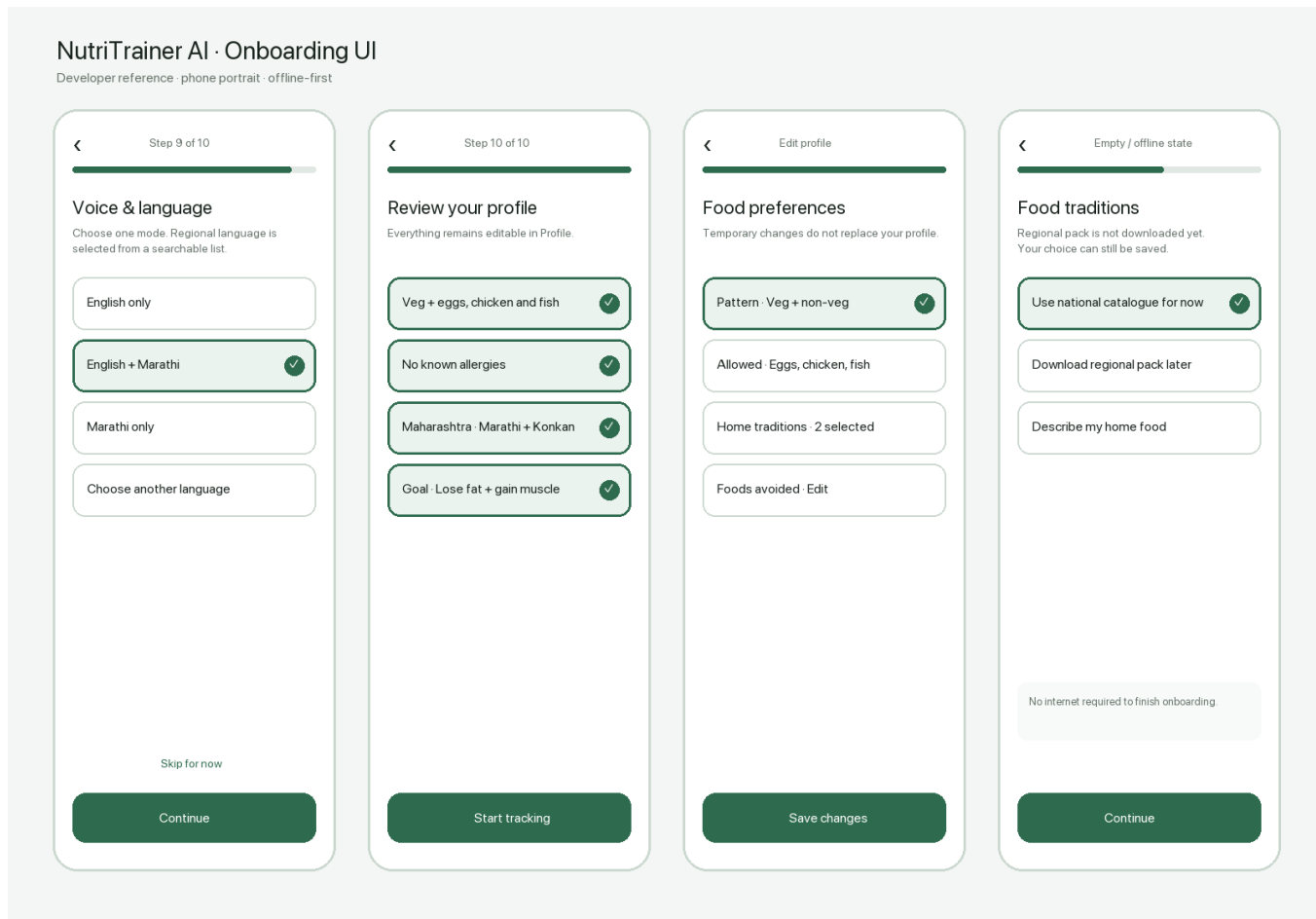


Figure 4 Voice/language, review, later editing and offline fallback state.

State model, events and acceptance criteria

OnboardingState fields: version, currentStep, completedStepIds, foodPattern, allowedAnimalFoods[], dairyAllowed, allergies, avoidances, countryCode, stateUtCode, subRegionIds, traditionIds, customTraditionText, householdProfile, primaryGoal, bodyProfile, appLocale, loggingLanguageCodes[], skippedFields, consentTimestamps and completedAt. Store stable IDs; labels come from versioned localization packs.

- Emit local privacy-safe events for screen_view, option_selected, search_used, skipped, validation_error, permission_result and onboarding_completed; never place free-text community, allergy or health values in analytics payloads.
- Given a clean install with no network, a user can complete onboarding and enter Today.
- No food recommendation incompatible with foodPattern, selected animal foods or allergy exclusions appears in onboarding preview or afterward.
- Veg + non-veg users can receive vegetarian suggestions at any meal; the selected animal-food list controls which non-veg suggestions are eligible.
- OB-03 permits several simultaneous selections and persists every selected stable ID.
- Body recomposition appears as “Lose fat + gain muscle” and maps to its own goal code.
- Logging language supports English only, English plus any supported regional languages, or regional-language-only configurations independently from the app display language.
- All 28 states and 8 union territories are searchable; no state is selected merely because it appears first alphabetically or because of the product team’s origin.
- Changing from veg plus non-veg to vegan clears or asks confirmation for incompatible animal-food details before saving.
- Screen reader, 200% text, keyboard/focus and back/resume tests pass on Android and iOS.

- A user can finish in under two minutes by selecting a food pattern, skipping optional regional details and choosing Track only.

4.2 Coach

- Persistent composer with keyboard, microphone, camera and barcode actions.
- Live speech transcript is editable before submission.
- Confirmation shows meal, status, estimated totals, confidence and Edit items.
- Corrections replace earlier values instead of adding duplicates.
- Just estimate resolves optional questions with lower confidence.
- Responses remain concise: understanding, result, observation and next action.

Voice flow

- User taps or holds the microphone and speaks.
- Platform speech recognition produces editable text.
- AI classifies intent and extracts foods and quantities.
- Resolver matches household defaults and identifies one material ambiguity.
- User confirms or answers one question.
- Record saves and Today updates immediately.

Supported intents

Intent	Example	Effect
Consumed	I had two fulka and dal.	Create consumed items.
Plan	I may eat dal dhokli.	Create planned items excluded from actual totals.
Correct	Rice was 120 g, not 200.	Replace quantity and recalculate.
Remove	Remove the peanuts.	Soft-delete with undo.
Repeat	Same breakfast as Tuesday.	Preview cloned items before saving.
Ask	Are my carbs too high?	No data mutation.
Suggest	What should I eat next?	Rank validated candidates; save only after selection.

4.3 Today

- Summary shows consumed calories, protein, carbohydrate, fibre and fat against targets.
- Planned values are separate and never inflate consumed totals.
- Timeline groups workout, meals, snacks, water and measurements.
- Each item reveals source, serving basis, confidence and range reason.
- Finalize day creates a snapshot while preserving editable revision history.

4.4 Progress

- Default to seven-day weight average.
- Show BMI, waist, body fat and muscle mass only when available.
- Compare BCA readings from the same device and similar conditions where possible.

- Explain consumer BCA and hydration variability.
- Every chart has an accessible text summary.

4.5 Library

- Search English, supported Indian scripts, transliteration, alias, brand and barcode.
- Indicate verified label, authoritative reference, community recipe, personal recipe or estimate.
- Store household recipe ingredients and cooked batch yield.
- Confirm exact product and package size before saving a barcode result.

5 Functional requirements

ID	Requirement	Priority
FR01	Accept text and voice input and preserve the original transcript.	Must
FR02	Classify log, plan, correct, remove, repeat, ask, suggest and finalize.	Must
FR03	Extract food, quantity, unit, preparation, meal and status into a strict schema.	Must
FR04	Resolve personal, household, local, regional and online sources in precedence order.	Must
FR05	Calculate ingredient, recipe and serving nutrition deterministically.	Must
FR06	Maintain separate consumed and planned totals.	Must
FR07	Provide revision history, undo and soft deletion.	Must
FR08	Show confidence, source and serving basis.	Must
FR09	Scan barcode and nutrition-label photograph.	Must
FR10	Operate core logging, calculations and history offline.	Must
FR11	Track workouts and optional wearable data.	Should
FR12	Track weight, BMI, measurements and BCA fields.	Must
FR13	Recommend culturally relevant validated meals.	Must
FR14	Require one food pattern; store multi-select food details, state/UT, sub-region, traditions and day overrides.	Must
FR15	Apply safety, food pattern, allowed-food, household and regional precedence to every search and recommendation.	Must
FR16	Cover all 28 states and 8 union territories with searchable aliases and expandable regional content packs.	Must
FR17	Provide encrypted backup export and import.	Should
FR18	Integrate Health Connect and HealthKit optionally.	Should
FR19	Support optional common Gemma engine later.	Could

Calculation rules

Calculation	Rule
Ingredient	Reference nutrients per 100 g × edible grams ÷ 100.
Recipe batch	Sum ingredients and explicit cooking fats; retain cooked yield.
Recipe serving	Batch nutrients × serving grams ÷ final cooked yield.
BMI	Weight kg ÷ height metres squared.
Daily total	Sum confirmed consumed items only.
Weight trend	Rolling seven-day average when sufficient data exists.
BCA comparison	Compare compatible fields and flag device or unit changes.

Household calibration

Foods begin with regional defaults. The app occasionally offers one-time calibration, such as weighing three cooked fulkas together. After confirmation, that household record becomes the default. Size, restaurant and added-fat modifiers create per-entry overrides.

Clarification policy

Ask when	Do not ask when
Raw versus cooked materially changes the result.	A saved default already resolves it.
Several products match with different nutrition.	The difference is immaterial.
Serving ambiguity changes today's advice.	The user selected Just estimate.
Allergen interpretation is uncertain.	The answer is already known and current.

6 Nutrition confidence and recommendation logic

Source precedence

Rank	Source	Use
1	User-confirmed label or recipe	Highest priority for that user and version.
2	Exact barcode or manufacturer label	Use after product and pack confirmation.
3	Licensed authoritative composition data	Generic ingredients and foods.
4	Curated regional recipe	Community default with assumptions.
5	Open community database	Validate and retain provenance.
6	Heuristic or LLM estimate	Last resort, ranged and low confidence.

Confidence bands

Band	Example	Presentation
Very high	Current barcode plus label photograph	Value with source.
High	Calibrated household serving	Value with tolerance.
Medium	Generic medium homemade fulka	Range; no interruption unless important.
Low	One bowl mixed sabji without preparation	Wide range and quick edit.
Unresolved	Product cannot be distinguished	Require selection before saving.

Data quality

- Store basis as per 100 g, per 100 ml, per serving or prepared.
- Keep kcal and kJ separate.
- Store label version, capture date and URL.
- Do not merge products by name alone.
- Detect implausible macros and serving sizes.
- Preserve original and normalized values.
- Version and roll back food database changes.

Recommendation engine

Shared deterministic rules filter and rank meals using remaining energy and protein, fibre range, diet, allergens, workout context, repetition, preparation time and regional preference. The LLM explains approved candidates but cannot invent nutrition or override safety filters.

7 Food database and open APIs

No single source covers Indian ingredients, regional recipes and branded products. Every imported record needs provenance, retrieval date, licence metadata and normalized nutrients. Publicly readable information is not automatically reusable in a commercial database.

Source	Coverage	Constraint	Planned use
Open Food Facts	Barcode API, product, ingredients, images and nutrition.	ODbL database; separate contents and image licences.	Primary open barcode lookup and contribution source.
USDA FoodData Central	Search and details APIs for foundation and branded foods.	API key; secure it and review redistribution terms.	Secondary generic and international reference.
GS1 Digital Link	Resolves GS1 identifiers to product information.	Product access varies by resolver/provider.	Identity and manufacturer-page discovery.
ICMR NIN IFCT 2017	528 Indian foods and extensive nutrients.	Not open API; electronic product use requires prior written permission.	Seek permission before ingestion.
FSSAI	Indian label regulations and standards.	Regulatory reference, not nutrition product API.	Validation and label interpretation.
Manufacturer pages	Current label when published.	Terms vary; scraping can be prohibited.	Controlled verification with capture date.

Open Food Facts plan

- Lookup GTIN before product-name search.
- Identify the application and request only required fields.
- Cache successful responses with source timestamp.
- Validate serving basis, units and macro totals.
- Display attribution and comply with ODbL, Database Contents Licence and image CC BY SA terms.
- Keep licence-sensitive sources separable until legal review confirms combination obligations.

USDA plan

- Use Foundation Foods for generic ingredients and branded data as a secondary source for India.
- Normalize nutrient identifiers while retaining FoodData Central IDs.
- Do not embed a private production API key in the application; use a minimal proxy if required.

Indian data plan

- Obtain written ICMR NIN permission before electronically storing IFCT data in a commercial product.
- Commission dietitian-reviewed recipes across launch regions with measured cooked yields.
- Moderate community submissions and retain recipe versions.
- Accept label photographs for missing products.
- Use FSSAI licence information as context, not proof of nutrition.

Unknown product workflow

- Scan barcode locally.
- Search personal cache, bundled catalogue and Open Food Facts.
- If online and unresolved, query controlled sources or manufacturer pages.

- Match barcode, brand, variant, flavour and package size.
- Capture front and nutrition panels with OCR when unresolved.
- Show extracted values for confirmation and save a user-verified version.

Source links

Open Food Facts product API <https://openfoodfacts.github.io/documentation/docs/Product-Opener/v3/products/get-api-v3-product-code/>

Open Food Facts licensing <https://openfoodfacts.github.io/documentation/docs/Product-Opener/api/tutorials/license-be-on-the-legal-side/>

USDA FoodData Central <https://fdc.nal.usda.gov/api-guide/>

GS1 Digital Link <https://ref.gs1.org/standards/digital-link/>

ICMR NIN IFCT https://www.nin.res.in/ebooks/IFCT2017_16122024.pdf

FSSAI labels https://www.fssai.gov.in/upload/uploadfiles/files/Comp_Labelling%20Display_Version%20VII_03042025.pdf

8 Local AI architecture

Component	Responsible for	Not responsible for
Speech recognition	Voice to editable text.	Nutrition meaning.
Language engine	Intent, entities, clarification and phrasing.	Calories, BMI, BCA math or history.
Resolver	Aliases, identity and household defaults.	Generating prose.
Calculation engine	Recipes, nutrients, totals and trends.	Guessing missing facts.
Recommendation engine	Filtering and ranking validated meals.	Diagnosis.
Local database	Authoritative user and food records.	Inference.

Engine strategy

Platform	Primary	Fallback
Android	Gemini Nano through ML Kit Prompt API when supported.	Deterministic parser; optional Gemma later.
iOS	Apple Foundation Models when available and locale-supported.	Deterministic parser; optional Gemma later.
Shared	Versioned NutritionIntent and response schemas.	Reject invalid output and request confirmation.

Migration

Every platform model implements NutritionLanguageEngine. A future Gemma E2B or E4B can replace or supplement system models without changing storage, calculations or screens. Current official mobile Gemma variants are E2B and E4B; no official E3B is listed as of this document.

Prompt contract

Send only the current utterance, compact daily summary, relevant household defaults and retrieved candidates. Require a schema containing intent, status, foods, quantities, units, modifiers, confidence and clarification. Invalid output is never written directly.

Quality gates

Measure	Threshold
Intent accuracy	At least 95% on release corpus.
Food and quantity exact match	At least 90% for high-frequency launch languages and English.
Plan versus consumed	At least 98%.

Measure	Threshold
Correction semantics	At least 98% without duplicate totals.
Schema validity	At least 99.5% after validation.
Unsupported inputs	No silent high-confidence fabrication.

Evaluation corpus

Create a consented or synthetic evaluation corpus across English and launch Indian languages/scripts, transliteration, code switching, typos, household units, raw/cooked ambiguity, products, corrections, plans and safety requests. Freeze a private regression set and run it after every prompt, API and OS change.

9 Recommended technology

Layer	Technology	Reason
Shared app	Kotlin Multiplatform	Shared models, calculations, repositories and validation.
Shared UI	Compose Multiplatform	One UI with native escape hatches.
Android AI	ML Kit GenAI Prompt API and AICore	Gemini Nano efficiency on eligible devices.
iOS AI	Apple Foundation Models	OS-managed local model on eligible devices.
Storage	SQLDelight or Room KMP with SQLite	Typed migrations and offline-first transactions.
Serialization	kotlinx.serialization	Versioned AI and database contracts.
Concurrency	Coroutines and Flow	Streaming voice, inference and UI state.
Network	Ktor Client	Shared HTTP, caching and retry.
Health	Health Connect and HealthKit	Permissioned health and fitness records.
Barcode	ML Kit Barcode and Apple Vision	Local recognition.
OCR	ML Kit Text Recognition and Apple Vision	Nutrition label capture.
Voice	Android on-device SpeechRecognizer and Apple Speech	Platform transcription with availability checks.
Testing	Kotlin tests, platform UI tests and golden corpus	Cross-engine regression.

Modules

Module	Contents
shared domain	Entities, units, nutrient ontology, calculations, goals and policies.
shared data	SQLite, migrations, repositories, API clients, caching and provenance.
shared AI contract	Model-neutral intents, validators and corpus runner.
shared UI	Compose screens, design system and state holders.
android platform	Gemini Nano, speech, barcode, OCR, Health Connect and keys.
iOS platform	Foundation Models, Speech, Vision, HealthKit and Keychain.
content pipeline	Imports, normalization, duplicate checks, signing and data packs.

Core entities

Entity	Key fields
UserProfile	units, height, goals, primary food category, allergies, language and current region.
FoodPreference	state/UT, sub-regions, food traditions, cuisines, non-veg details, exclusions and schedule.
DietaryOverride	date, fasting rule, vegetarian or vegan day, egg rule and expiry.
RegionTag	country, state/UT, sub-region, aliases, local scripts and content-pack version.
Food	canonical name, aliases, nutrients, basis, source, confidence, category tags and regions.
Product	GTIN, brand, variant, pack, label version, ingredients and nutrients.
Recipe	ingredients, preparation, raw weights, cooked yield, category tags, regions and owner.
MealItem	meal, status, food, quantity, modifiers, range and revision.
Workout	type, duration, intensity and source.
Measurement	weight, waist, date, conditions and source.
BodyComposition	device, fat, muscle, visceral metric and original report.
DailySummary	consumed and planned totals plus calculation version.

Offline, backup and security

- SQLite remains the local source of truth.
- Queue idempotent network work and cache food lookups.
- Offer encrypted export/import before managed cloud synchronization.
- Never embed private production API keys in the binary.
- Sign downloadable food and model packs.
- Exclude health content from diagnostic logs and default analytics.
- Provide complete export, deletion and permission revocation.

10 Privacy safety accessibility and compliance

Privacy requirements

Area	Requirement
Consent	Explain local and online processing when enabled.
Microphone	Record only during explicit listening with a visible stop control.
Photos	Process locally when possible; consent before uploading.
Health	Request only fields required for enabled features.
Training	Never train on personal data without separate opt-in.
Deletion	Remove history, cached images, backups and personalization.
Children	Do not launch for minors before age-specific design and consent.

Safety

- No diagnosis, medication advice or unsafe calorie restriction.
- Escalate eating-disorder signals, severe symptoms and urgent concerns.
- Use allergies as hard filters while disclosing incomplete data and cross-contamination risk.
- Pregnancy, diabetes, kidney disease and clinical contexts receive conservative guidance and professional referral.
- Never present consumer BCA as clinically exact.
- Separate informational ranges from personalized medical targets.

Accessibility and localization

- Platform-size touch targets and accessible labels.
- Never encode confidence through colour alone.
- Dynamic text without clipped cards or nutrition rows.
- Charts include text summaries and selectable values.
- Voice provides transcript review, cancel and keyboard alternative.
- Preserve local script and transliteration under language-neutral IDs.
- Support grams, millilitres, household units, kilograms and pounds.

Legal review

- Open Food Facts ODbL compliance and source separation
- Product-image attribution and trademarks
- IFCT written permission
- USDA and third-party terms
- HealthKit and Health Connect policies
- India Digital Personal Data Protection obligations
- Wellness claims and store health declarations

11 Analytics testing and operations

Privacy-preserving events

Event	Allowed	Excluded by default
meal log completed	input mode, duration and confidence band	Transcript and exact foods.
clarification shown	reason category	Question text.
AI engine selected	engine, availability and latency	Prompt and response.
lookup completed	source, hit or miss and latency	Unprotected barcode.
error	coded failure and app version	User text, images and health data.

Testing matrix

Area	Required tests
Calculations	Golden arithmetic, units, yields, rounding and ranges.
Conversation	Intent corpus, plans, corrections and ambiguity.
Database	Migrations, offline queue, edits and recovery.
AI engines	Same corpus on every model and supported OS.
Voice	Noise, accents, transcript edits and permission denial.
Images	Rotation, glare, multilingual and serving bases.
Accessibility	Screen reader, dynamic type, contrast and reduced motion.
Performance	Low, mid and high-tier devices; cold and sustained use.

Operational measures

- Model cold start, warm latency and tokens
- Speech latency and correction rate
- Food lookup hit rate by source and region
- Low-confidence rate by category
- Database migration and rollback
- Battery, memory and thermal behaviour
- Crash-free and application-not-responding rates

12 Delivery plan and acceptance

Phase	Scope	Exit
Discovery 3 to 4 weeks	Naming, interviews, regional corpus and AI spikes.	Validated loop, device matrix and licensing path.
Prototype 4 to 6 weeks	Coach, Today, text, 100 foods and calculations.	End-to-end logs and corrections pass tests.
Private alpha 6 to 8 weeks	Voice, both platforms, recipes, barcode, OCR and progress.	90% successful interpretation with 30 testers.
Beta 6 to 10 weeks	Open Food Facts, health integration, export and content tools.	Performance, privacy and store readiness.
Launch	India-wide taxonomy with quality-gated regional packs.	Quality gates sustained.
Later	Optional Gemma, more content packs, sync and partnerships.	Independent evaluation and licence approval.

MVP

- Voice and chat logging
- Coach and Today
- Consumed, planned and corrected states
- Local foods and household defaults
- Deterministic calories and macros
- Weight and BMI trends
- India-wide region selection and regional suggestions
- Barcode and label capture
- System LLM plus parser fallback
- Editable history, source and confidence

Definition of done for a meal

- Original input retained.
- Intent and items pass schema validation.
- Every food resolves to a source or explicit estimate.
- Consumed state is correct.
- Calculation version and confidence saved.
- Coach and Today totals match.
- Edit, undo and delete work.
- Cached-food path works offline.

13 Risks and open decisions

Risk	Mitigation
System LLM unavailable	Capability check, parser fallback and optional Gemma.

Risk	Mitigation
Cross-model inconsistency	Strict schemas, shared validation and frozen corpus.
Regional language weakness	Aliases, transcript editing and language-specific evaluation.
Database licence exposure	Separate sources, counsel and NIN permission.
False precision	Ranges, confidence and serving basis.
Health advice harm	Scope limits, deterministic guards and escalation.
Missing products	Barcode, OCR, user verification and moderation.
Data loss	Encrypted export first; optional backup later.
OS model regression	Version prompts and continuously retest.

Open decisions

Decision	Recommendation
Name	NutriTrainer AI selected; complete trademark, domain and app-store clearance.
Launch focus	India-wide taxonomy with quality-gated regional and cultural content packs.
Audience	General adult wellness with strength and fat-loss journeys.
Backup	Encrypted export first, managed sync after validation.
Food data	Pursue NIN permission while building separable open catalogue.
Fallback model	Parser for MVP; benchmark Gemma E2B before beta.
Business model	Decide after retention and operating evidence.

Immediate next steps

- Approve product boundary and MVP.
- Choose launch languages and regional content-pack sequence.
- Specify nutrient ontology and calculation rules.
- Create a multilingual regional evaluation corpus.
- Prototype both system LLMs behind one contract.
- Contact ICMR NIN about IFCT permission.
- Test Coach-to-Today corrections with 10 target users.
- Benchmark KMP storage, speech and inference on representative devices.

Appendix A Example behaviour

User says	Expected behaviour
I had one fulka.	Use household default or medium regional range.
I may have 200 g rice.	Create planned item and projected total.
Final rice was 120 g.	Replace plan with consumed quantity.
Same breakfast as Tuesday.	Preview copied items before saving.
This bread has 12.8 g protein per two slices.	Create a user-confirmed label version.
Just estimate it.	Use best range and stop questioning.
Why did weight rise overnight?	Use trend context, not instant fat-gain claims.
Compare my BCA.	Compare compatible fields and disclose variability.

Appendix B Architecture references

Gemini Nano <https://developer.android.com/ai/gemini-nano>

ML Kit Prompt API <https://developers.google.com/ml-kit/genai/prompt/android/get-started>

Apple Foundation Models <https://developer.apple.com/documentation/FoundationModels>

Gemma mobile <https://ai.google.dev/gemma/docs/integrations/mobile>

LiteRT LM <https://github.com/google-ai-edge/LiteRT-LM>

Kotlin Multiplatform <https://kotlinlang.org/docs/multiplatform/kmp-overview.html>

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